

Mini Ant Robot for SAR



ME 1670 Introduction to Engineering Graphics and Visualization

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ME 1670: Introduction to Engineering Graphics

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Chapter 1

Project Ideation

Project Description

Natural disasters inevitably take place around the world, and certain buildings cannot withstand the impact. When these buildings collapse, many humans may become trapped with little hope for escape. We hope that our Mini Ant Robot can locate and aid in the rescue of these victims. For movement, we took inspiration from small creatures, like spiders and worms. We plan to incorporate small spider legs in the front and rear to help the robot maneuver over and around fallen objects. Additionally, treads will be added to the sides of the robot so that our robot can slither through the wreckage like a worm. The frame will be made of flexible materials to resist the strain of tight spaces, and there will be an easily replaceable battery to sustain the robot's impact on the move. As for recognizance, we plan to mount a mobile camera spinning about an axis on the top of the robot. The mobility of this camera should provide the user with a 360-degree view of the wreckage. Additionally, our Mini Ant Robot will be equipped with multiple LED lights to help with navigation.

Constraints and Requirements

The three main constraints for designing a Mini Ant Robot as described above include:

- The robot must be small enough to fit through the gaps between rubble yet must be compact full of equipment to be functional
- The robot must have flexible frame and a suspension system to protect the electronics and absorb pressure when fitting through tight spaces
- The robot must be able to quickly switch out batteries to stay operational despite a limited power supply

Additionally, the Mini Ant Robot must have the mechanical capabilities to:

- Crawl over large rubble
- Slide through narrow gaps
- Relay visual information to its user
- Alert those trapped in the rubble

Description of Geometric Features

Camera Subassembly (Caden Whalen)

The camera subassembly is responsible for relaying visual information of the wreckage to its user. To keep the costs of the camera system low, a 360-degree view of the environment can be created with the help of two servo motors with one controlling horizontal movement and the other controlling vertical movement.

Battery Subassembly (Caden Whalen)

The battery subassembly is responsible for enabling its user to prolong the effectiveness and utility of the Mini Ant Robot. With a quick and simple push-push mechanism, users can quickly replace batteries on the move.

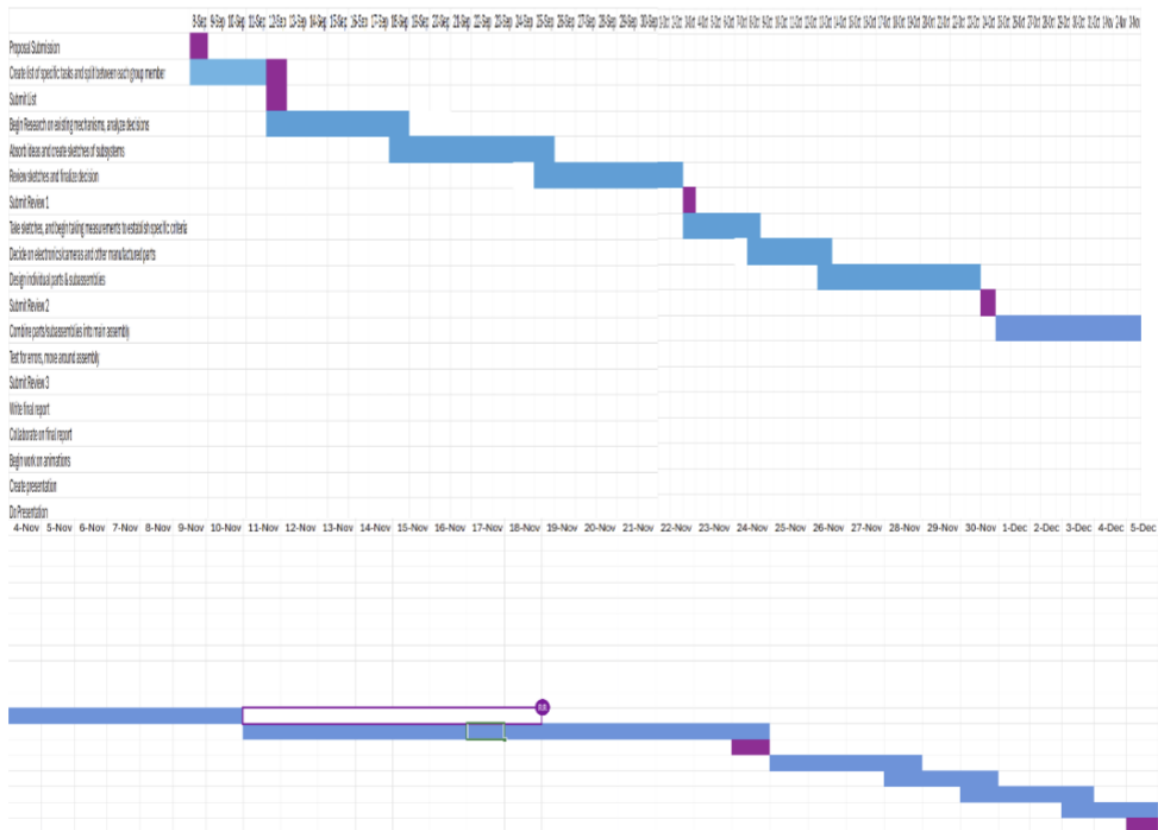
Suspension/Treads Subassembly (Rayyan Rahman)

The suspension and treads of the Mini Ant Robot are extremely important to its success because it enables the robot to slither horizontally and absorb stress. With a functional suspension system, the robot can maintain contact in uneven terrain and prevent internal damage. The treads make sure that the robot can maintain traction in harsh environments.

Spider Leg Subassembly (Luke Tumpalan)

The spider legs are responsible for the vertical mobility of the Mini Ant Robot. The spider leg subassemblies are meant to work together to elevate the robot when faced with large rubble that the treads cannot overcome.

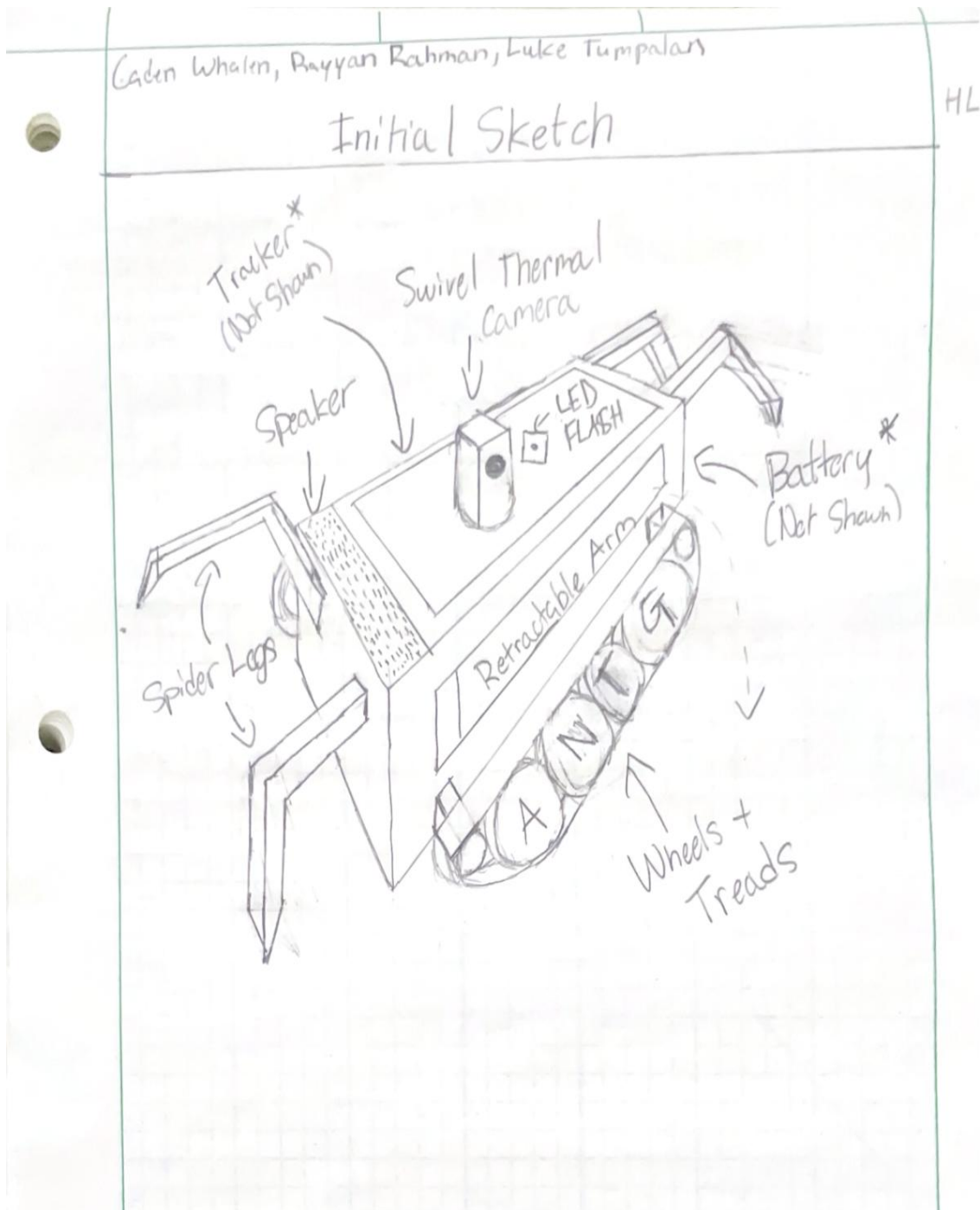
Project Management, Planning, and Timeline



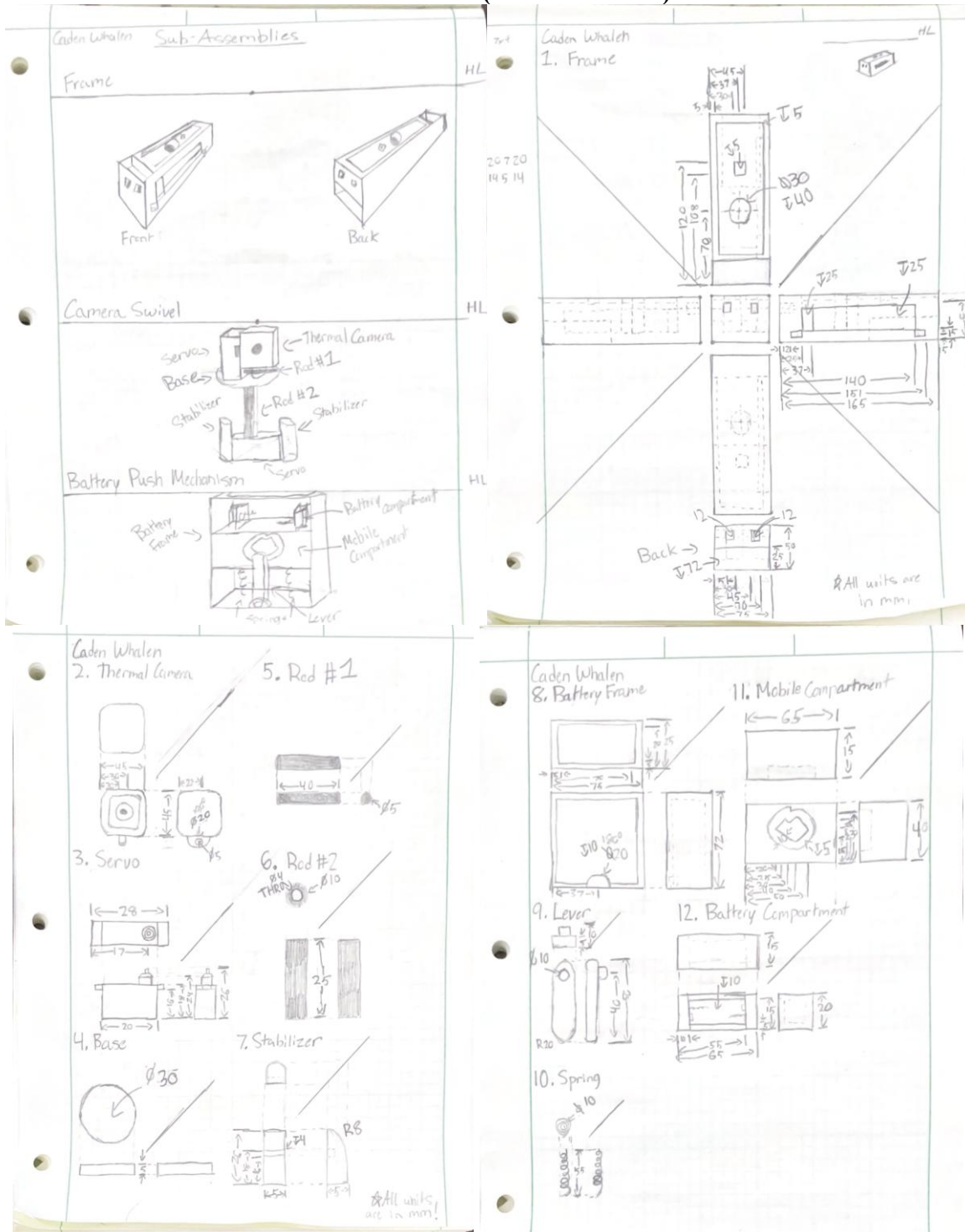
Chapter 2

Preliminary Design

Initial Perspective Sketch

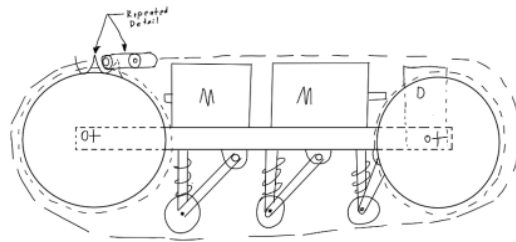


Hand Sketches (Caden Whalen)

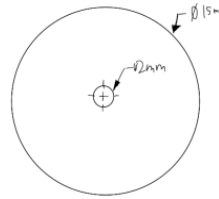


Hand Sketches (Rayyan Rahman)

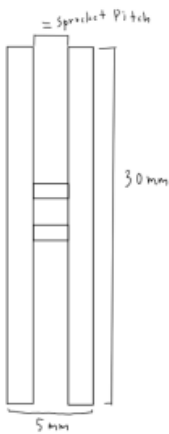
Sub Assl



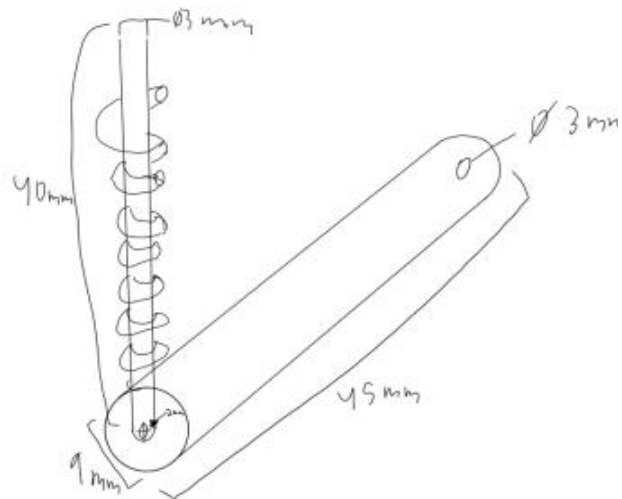
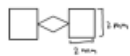
Caster



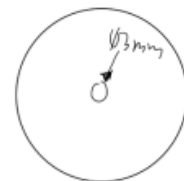
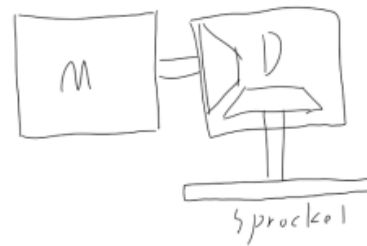
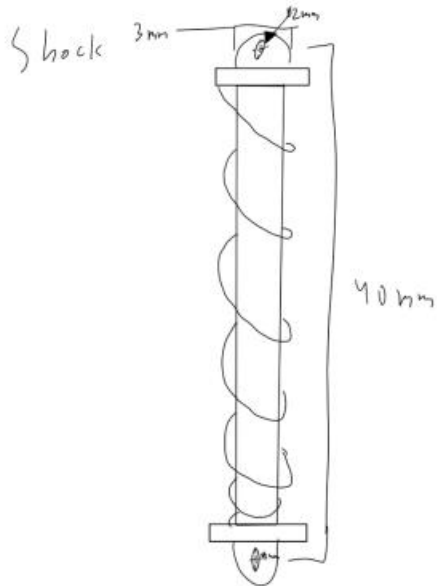
Tank Tread



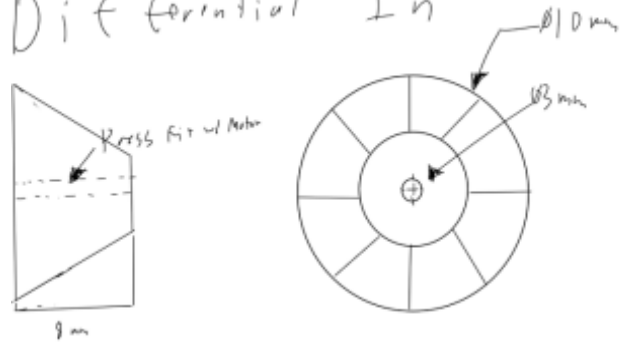
Rock on Subassl



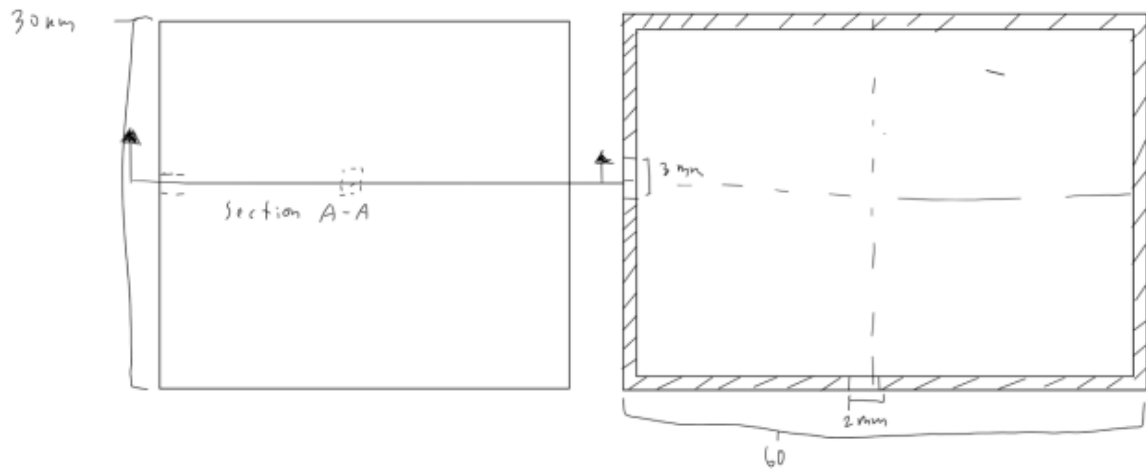
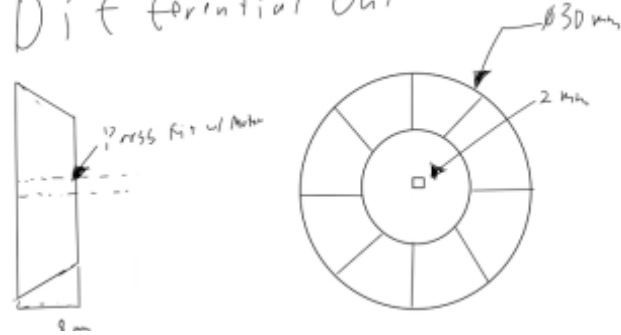
Drive System



Differential In

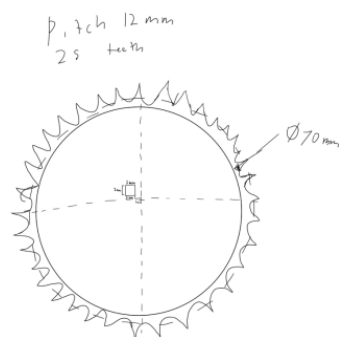
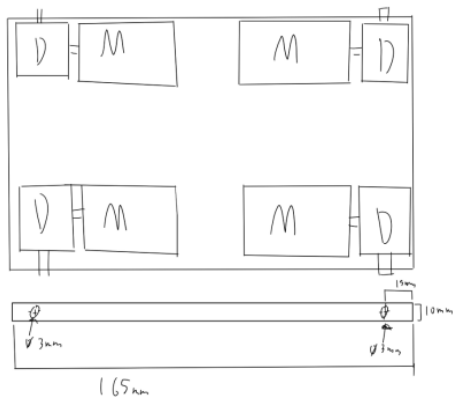


Differential Out



Frame

Sprocket



[illegible]

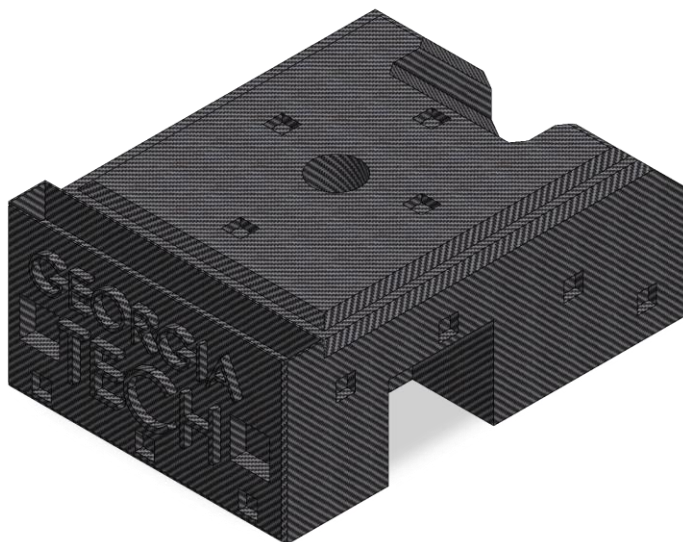
Chapter 3

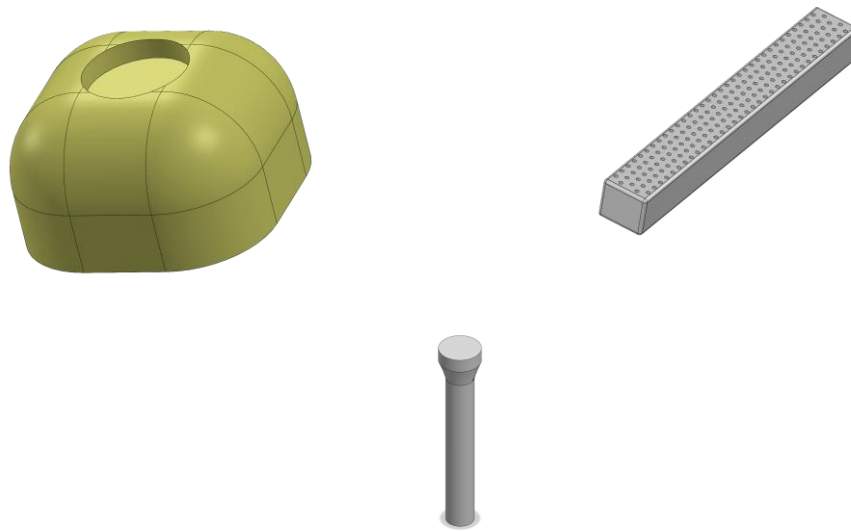
Detail Design

Final Assembly

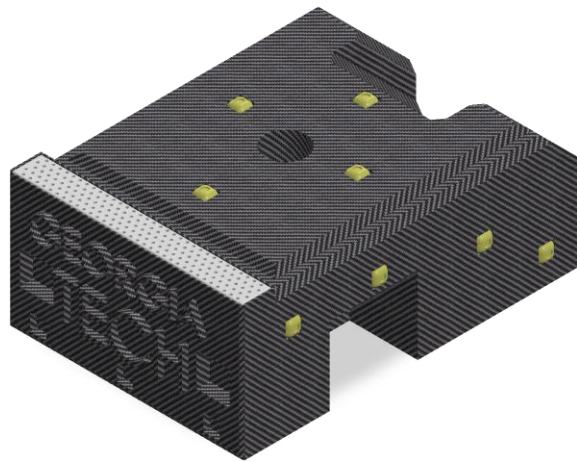


Frame Parts (Caden Whalen)



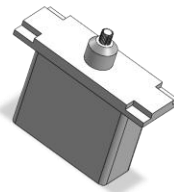
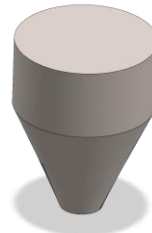
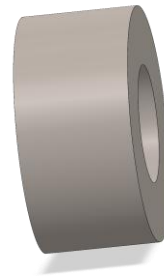
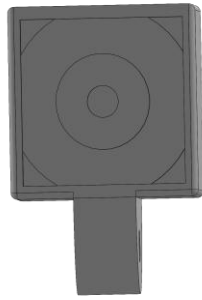


Frame Subassembly (Caden Whalen)

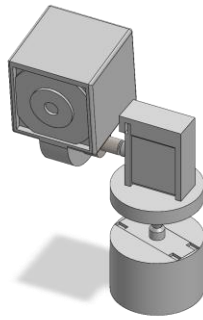


Camera Parts (Caden Whalen)

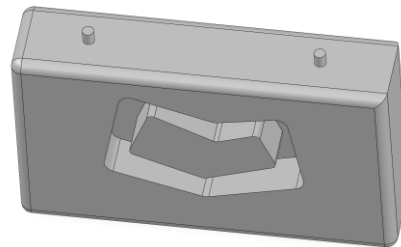
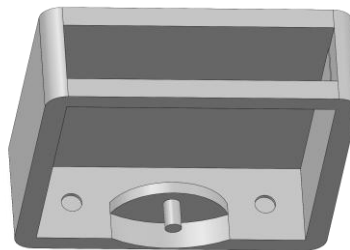
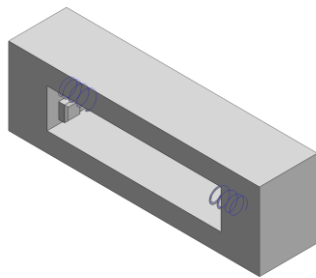




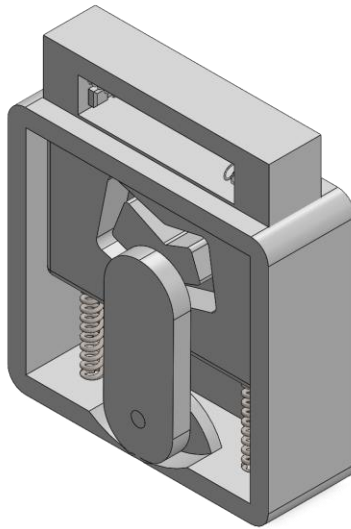
Camera Subassembly (Caden Whalen)



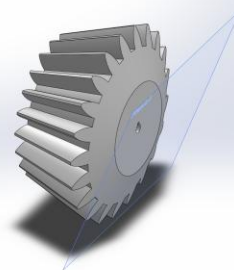
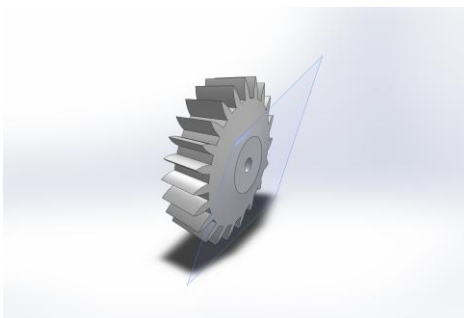
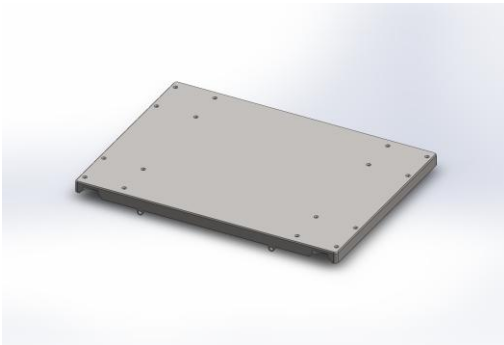
Battery Parts (Caden Whalen)

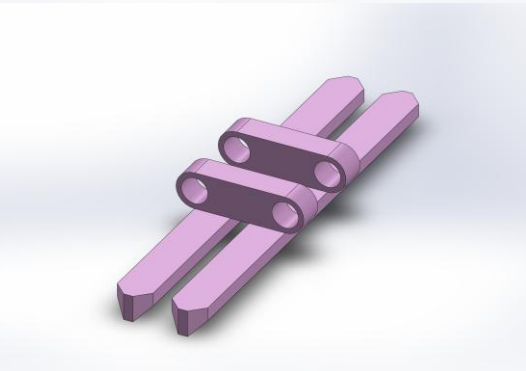
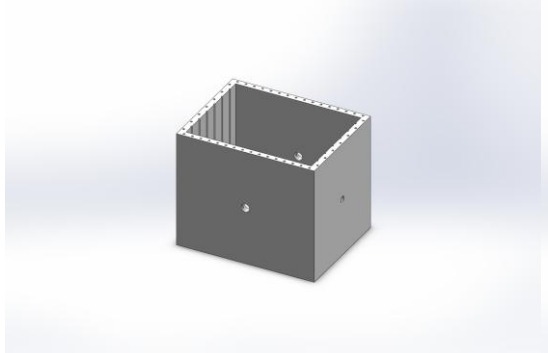
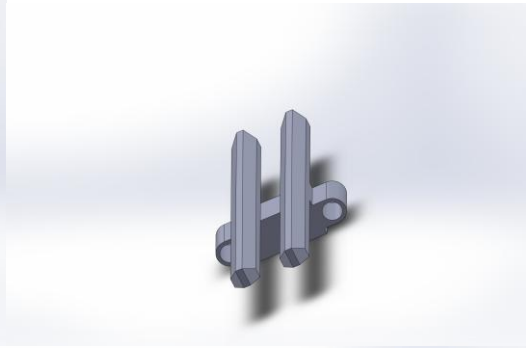
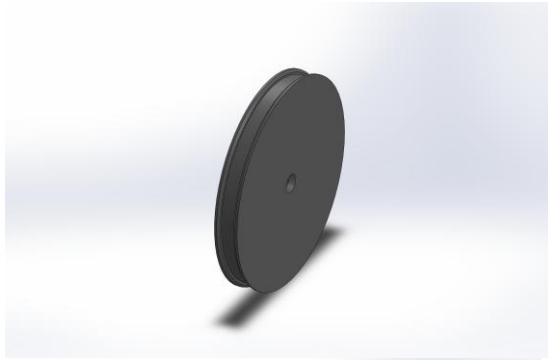


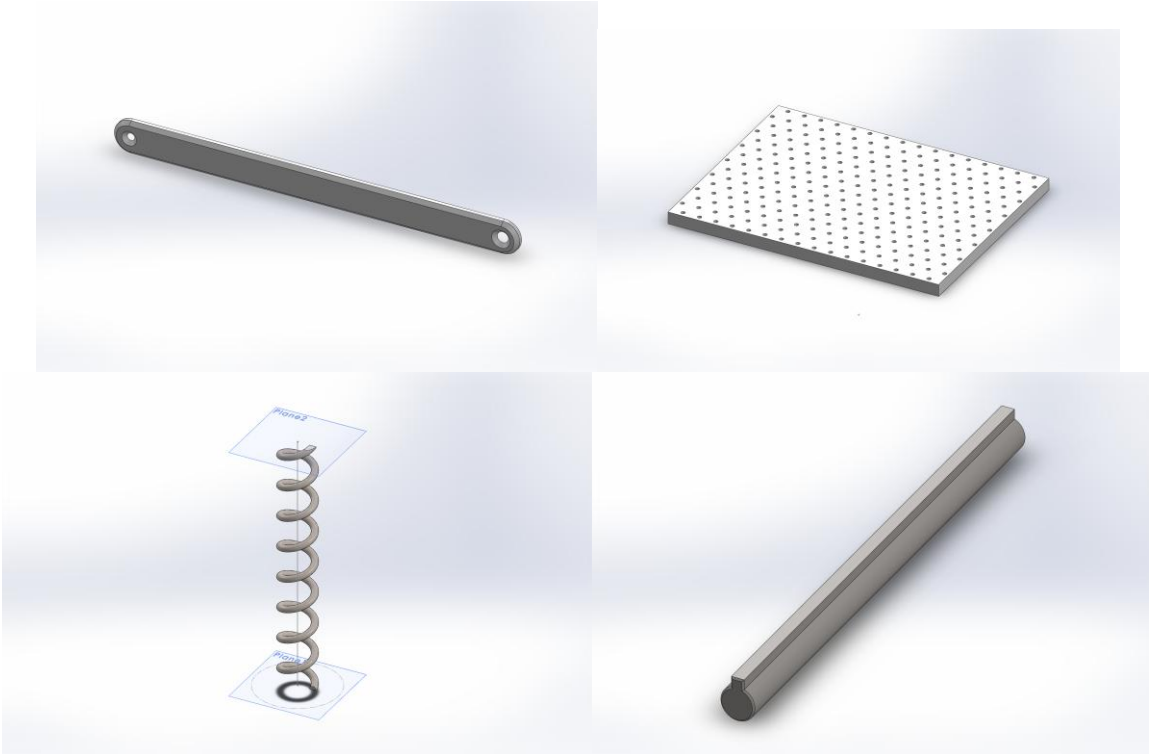
Battery Subassembly (Caden Whalen)



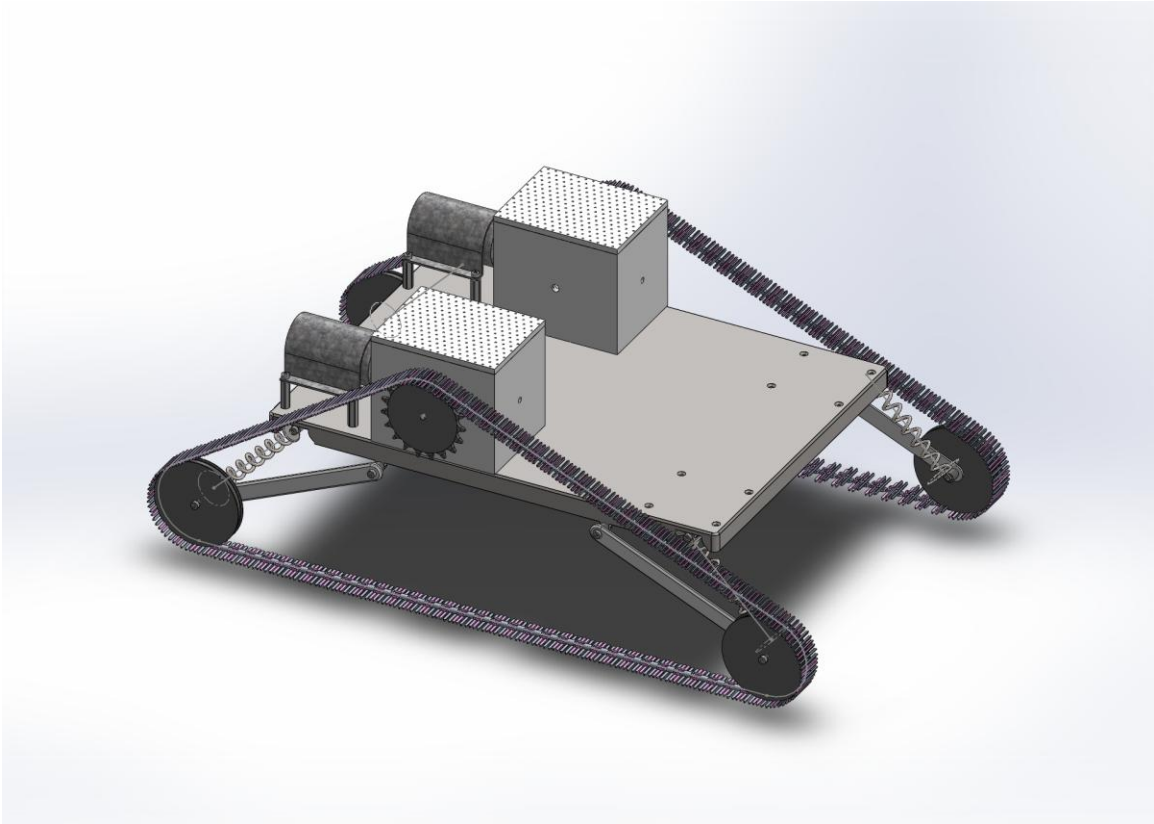
Suspension/Treads Parts (Rayyan Rahman)



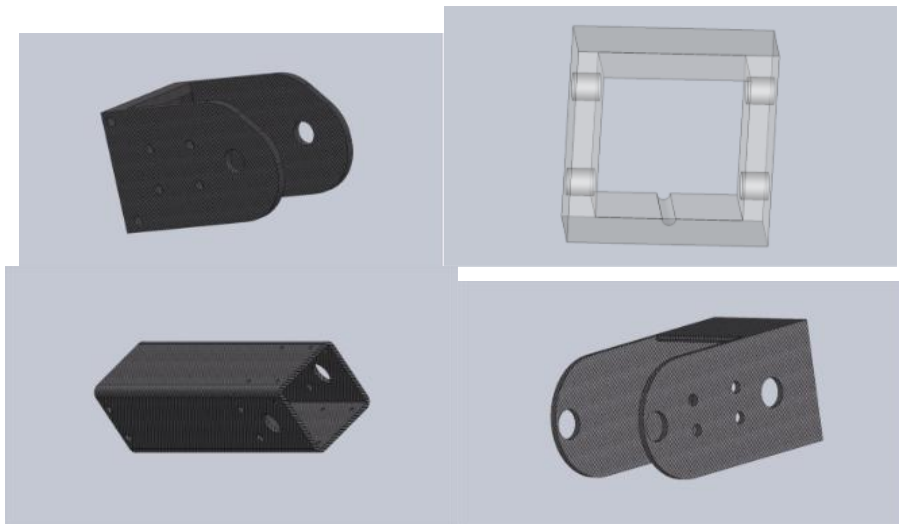


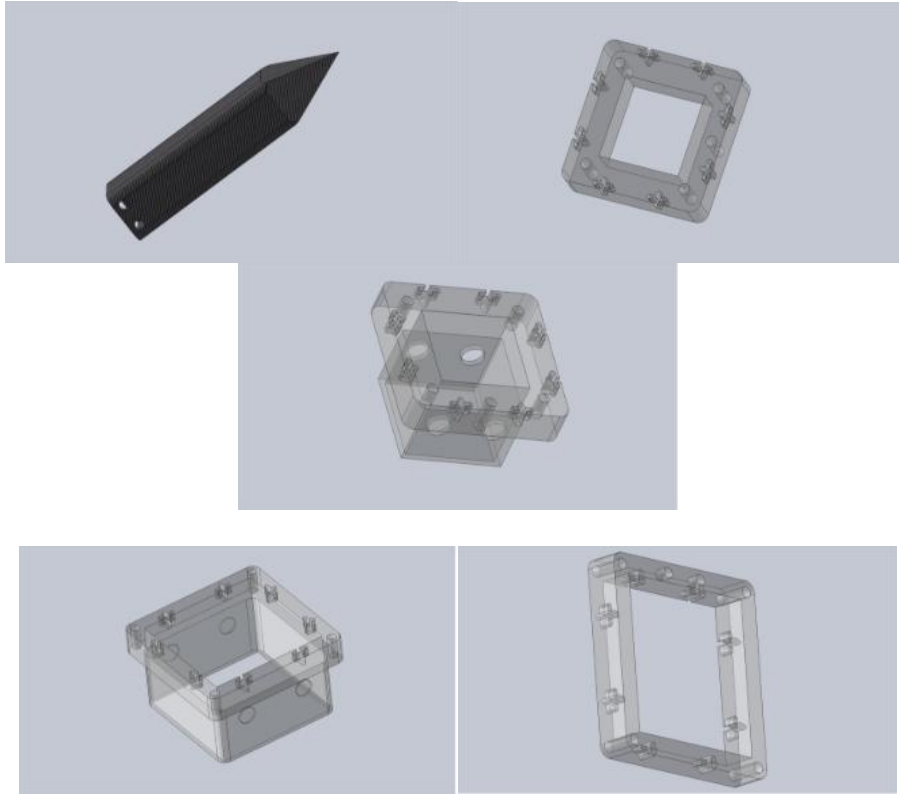


Drivetrain/Suspension Subassembly (Rayyan Rahman)



Spider Leg Parts (Luke Tumpalan)



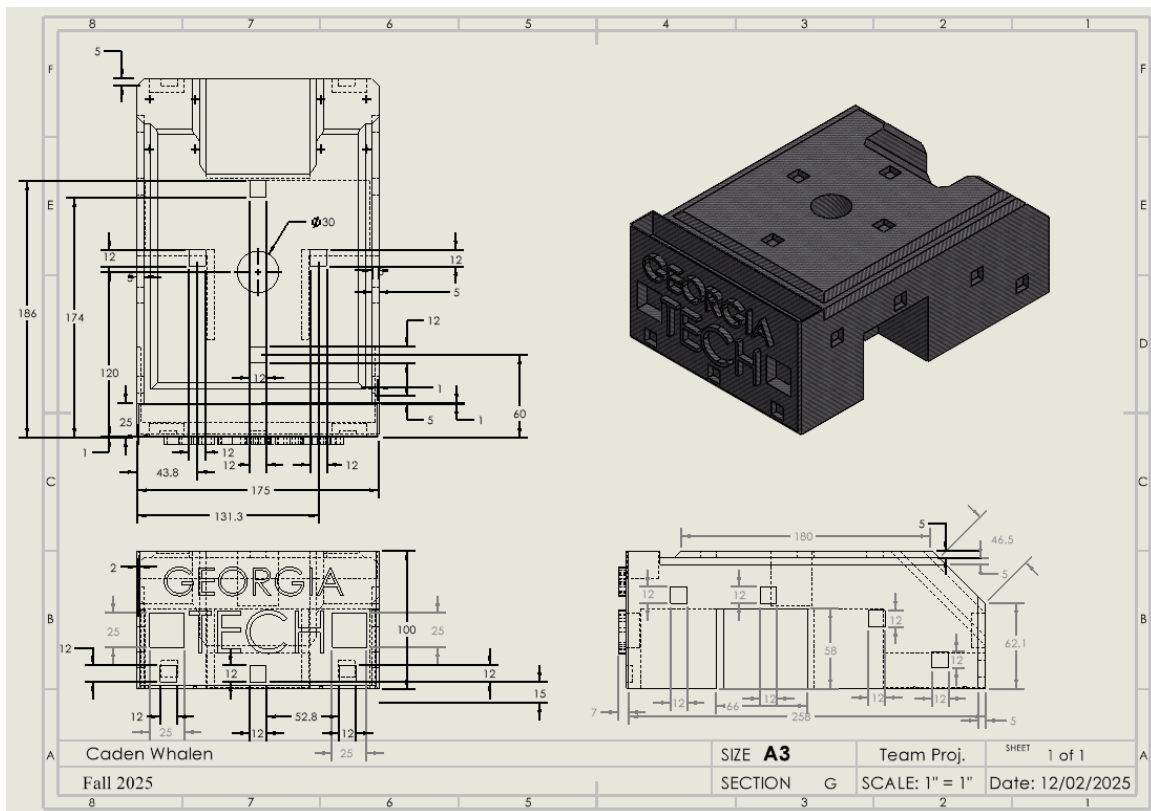


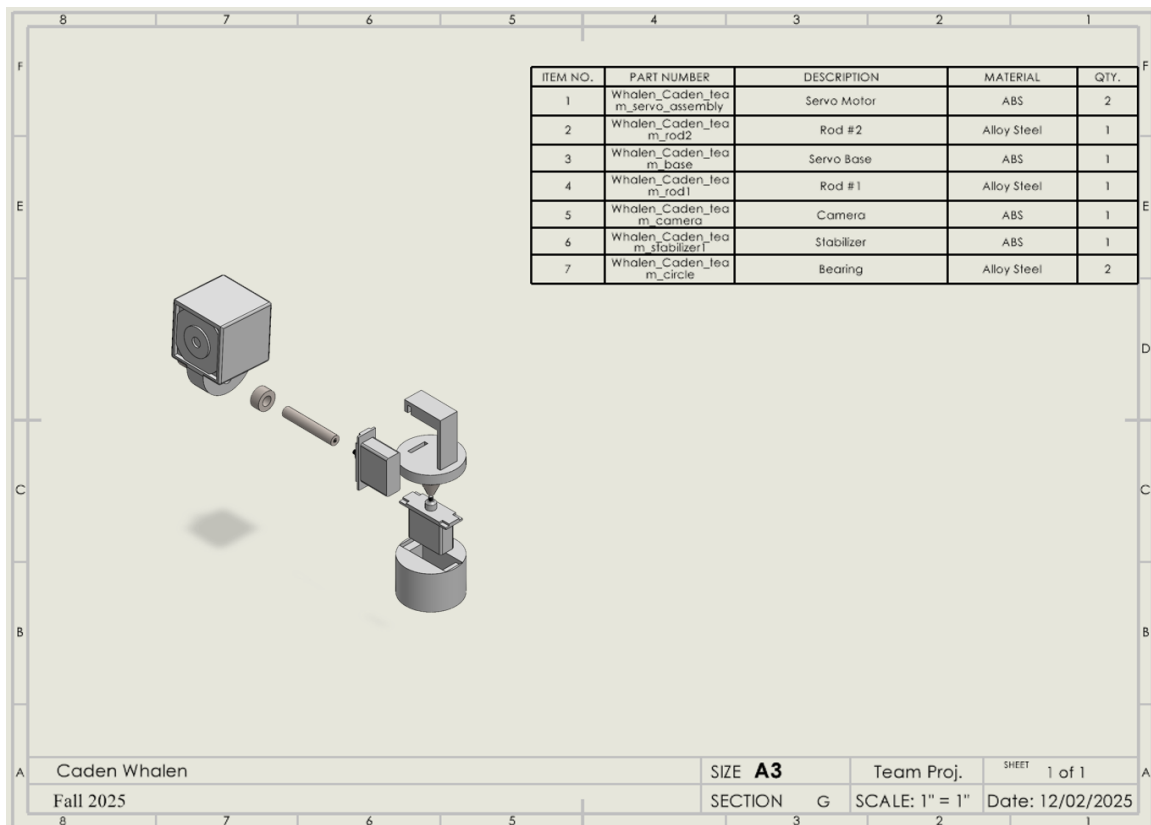
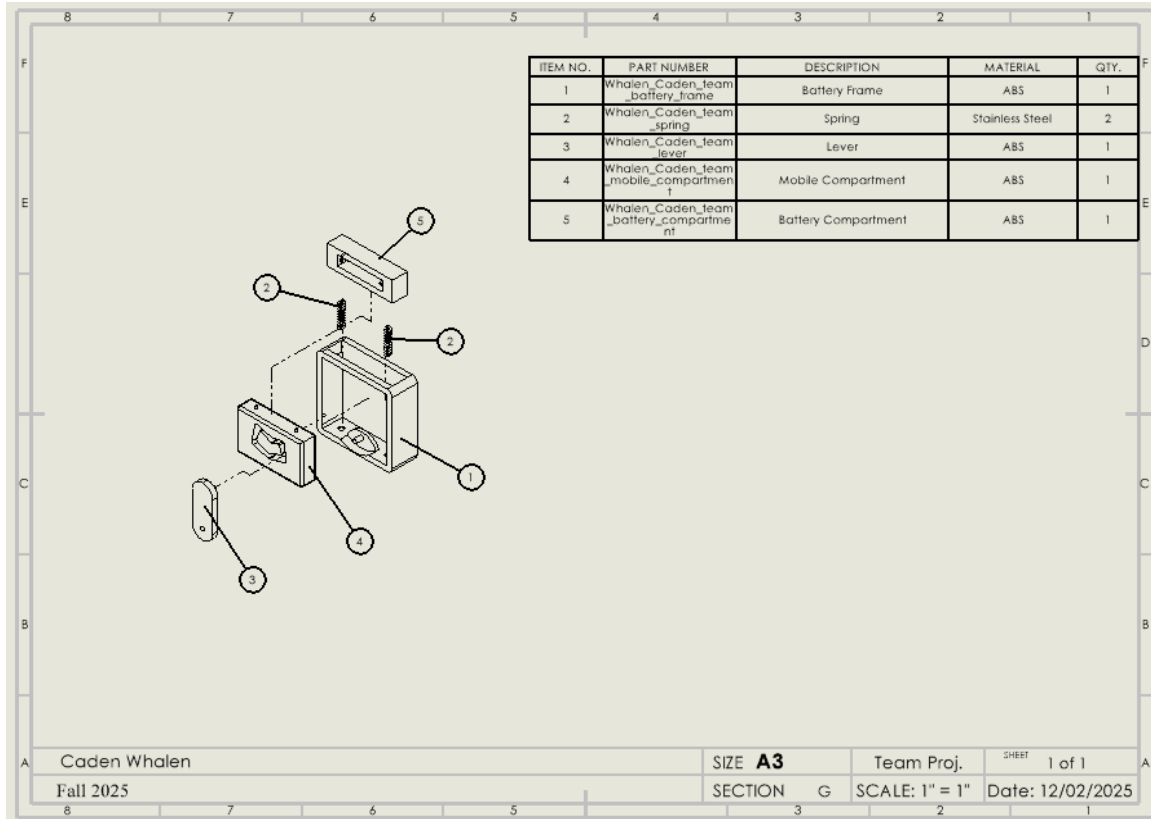
Spider Leg Subassembly (Luke Tumpalan)



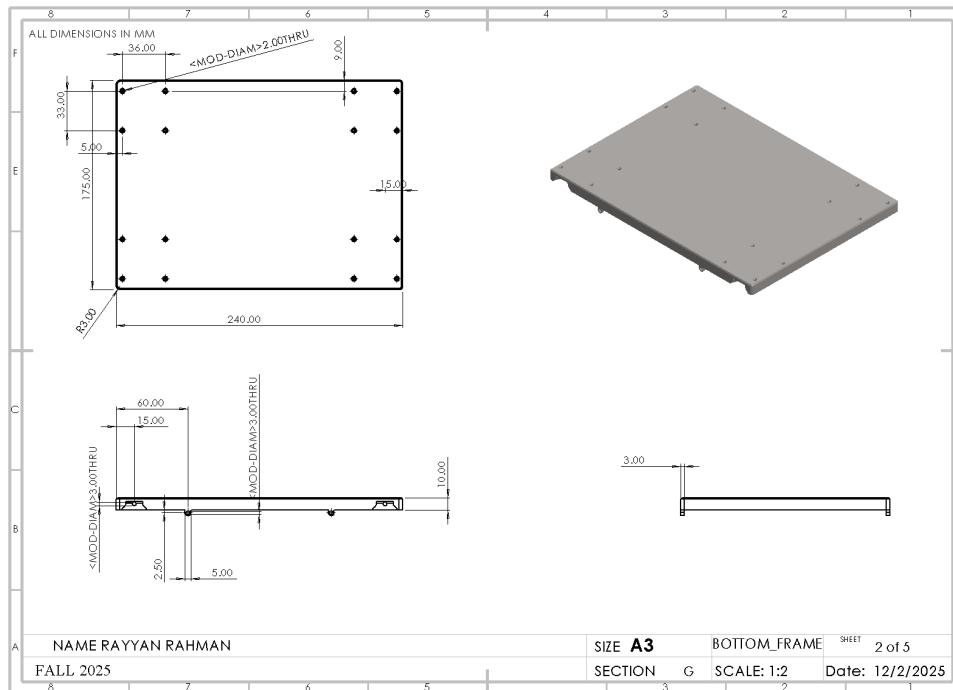
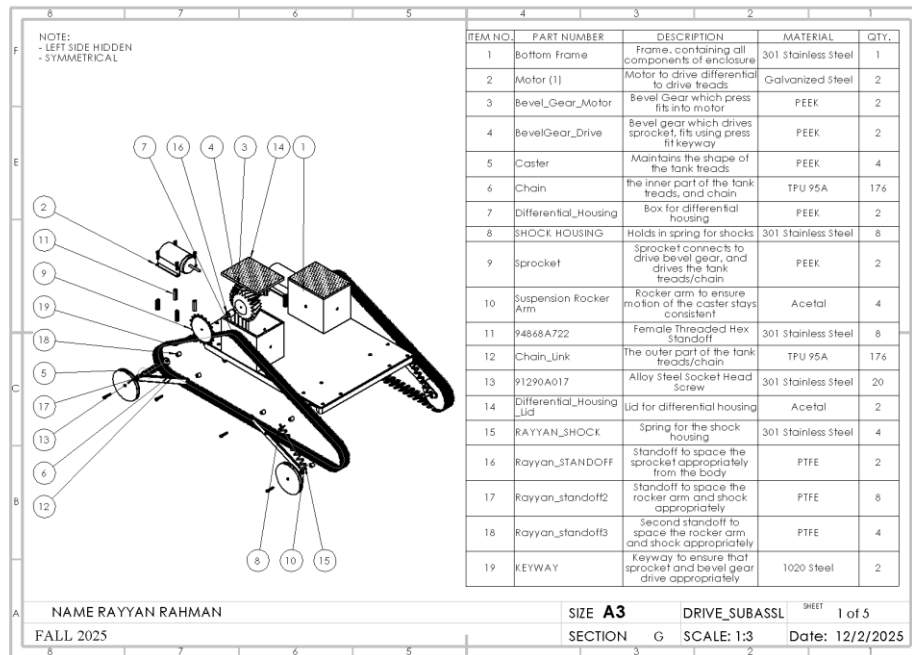
Manufacturing Working Drawings

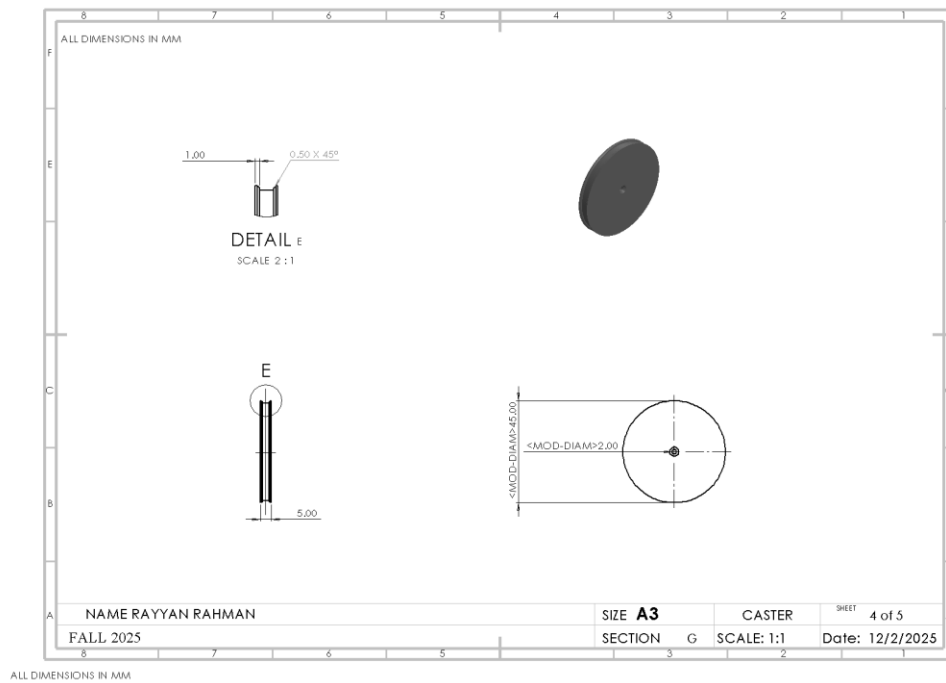
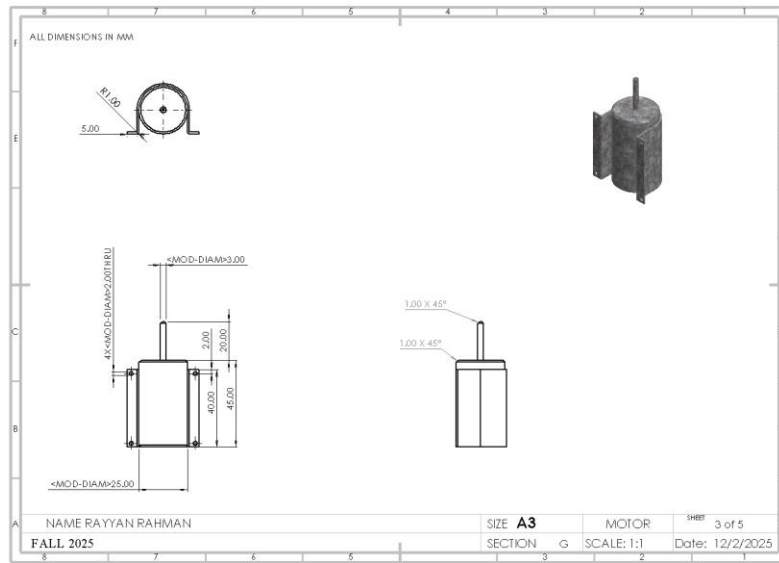
SolidWorks Drawings (Caden Whalen)

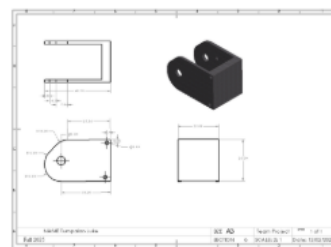
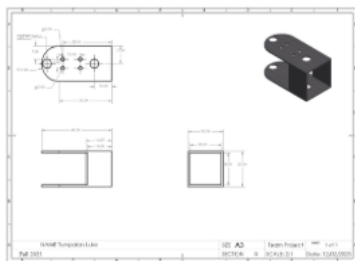
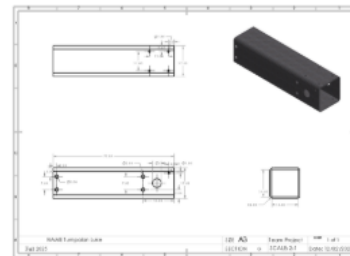
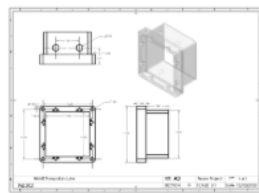
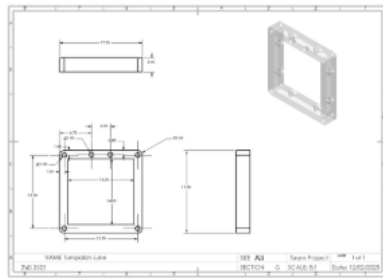
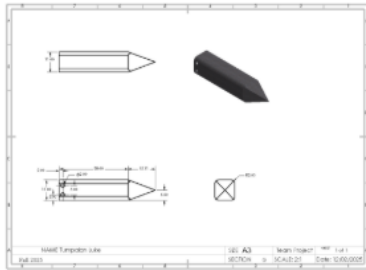
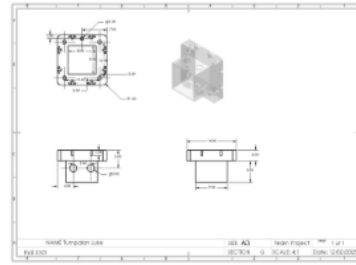
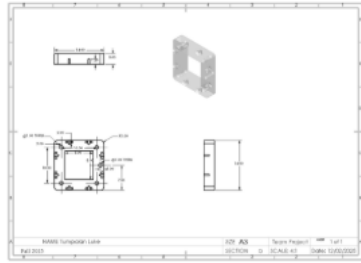


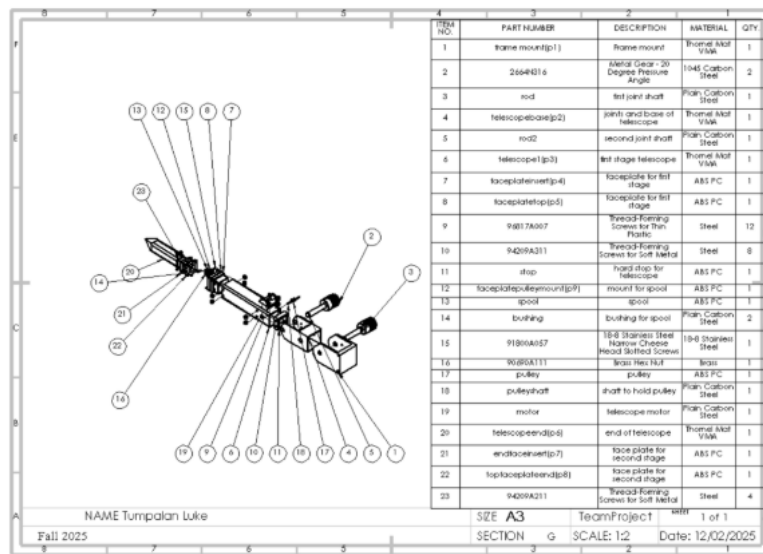


SolidWorks Drawings (Rayyan Rahman)









Frame Subassembly Instructions (Caden Whalen)

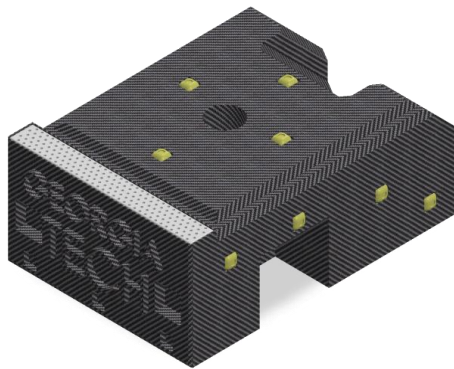
Step 1: Stick the 17 LED lights onto the robot frame by peeling off the backing and pressing the sticky side onto the cutout for at least 30 seconds



Step 2: Slide the speaker into the speaker notch on the top of the frame.

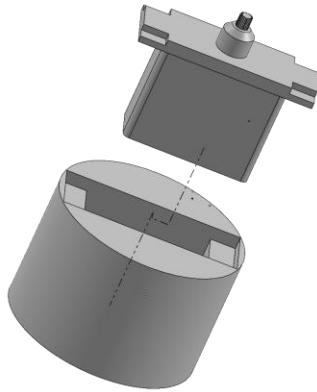


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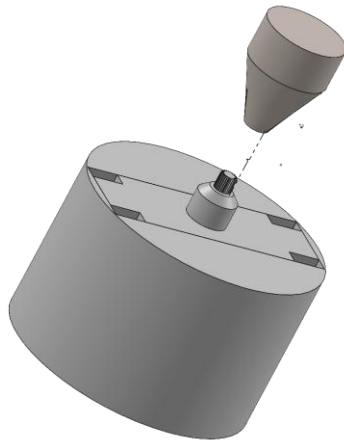


Camera Subassembly Instructions (Caden Whalen)

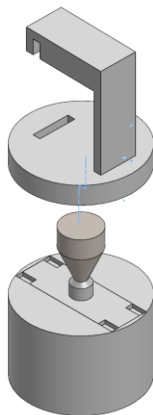
Step 1: Slide the lower servo motor into its stabilizer



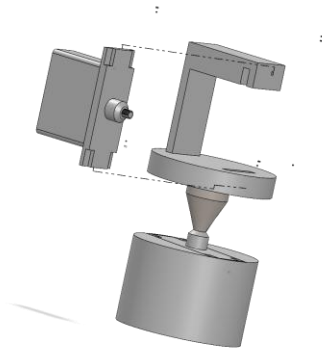
Step 2: Attach rod #1 onto the top of the servo motor and twist to ensure a good fit.



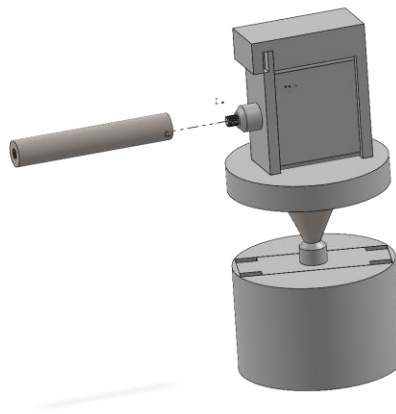
Step 3: Attach the base of the upper servo motor to the top of rod #1.



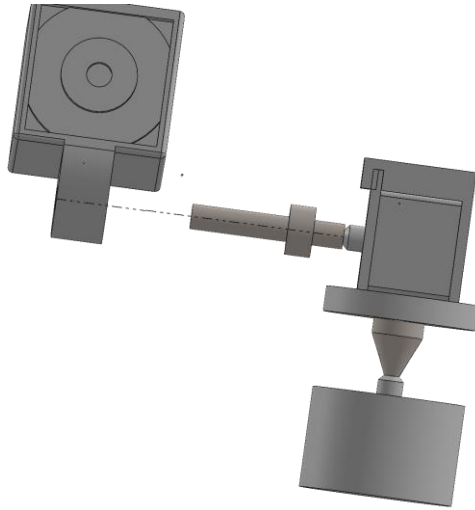
Step 4: Slide the upper servo motor into the slots of the camera base.



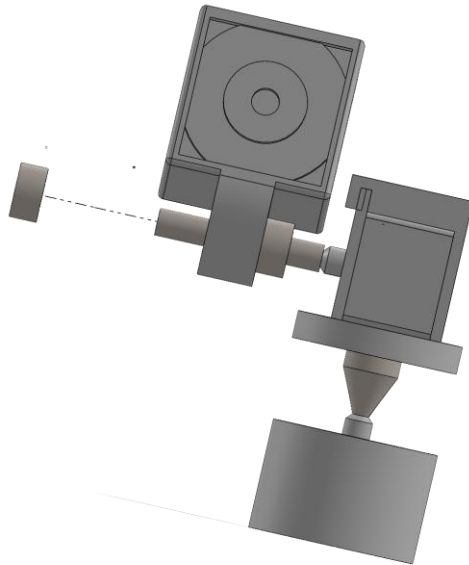
Step 5: Attach rod #2 to the upper servo motor and twist to ensure a good fit.



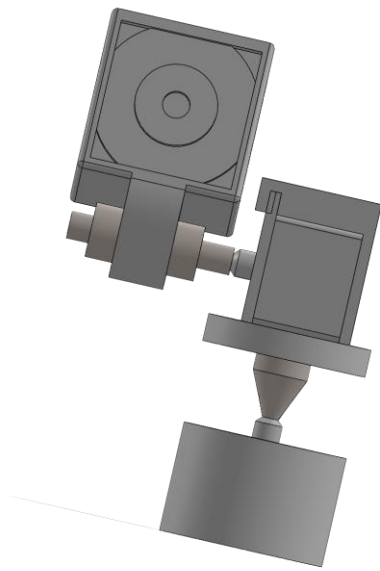
Step 6: Slide the first bearing and the camera halfway onto rod #2 and twist the first bearing to ensure limited movement.



Step 7: Slide the second bearing onto rod #2 until it is in contact with the camera. Make sure to tighten the second bearing to ensure limited movement and a tight fit.

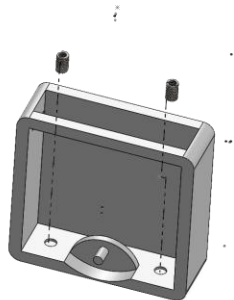


Final:

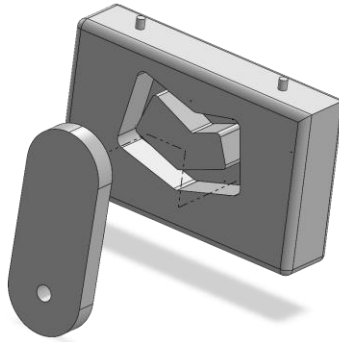


Battery Subassembly Instructions (Caden Whalen)

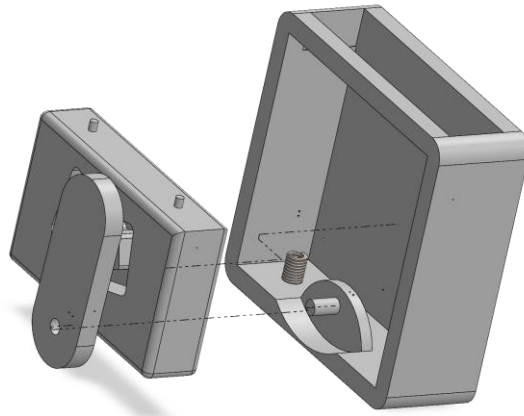
Step 1: Press the two springs into the circular cutouts located on the bottom of the battery frame.



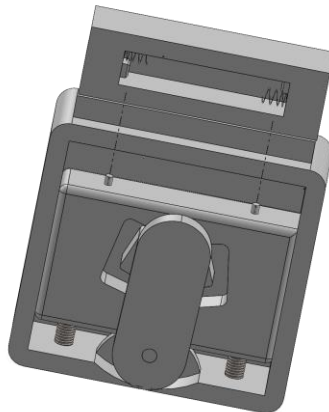
Step 2: Attach the tip of the lever into the cutout path of the mobile compartment.



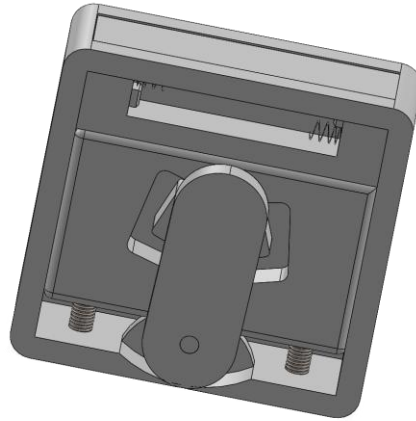
Step 3: Slide the lever and the mobile compartment into the battery frame. Make sure that the circular cutout in the lever aligns with the circular extrusion in the bottom of the battery frame.



Step 4: Attach the battery compartment to the top of the mobile compartment as shown.

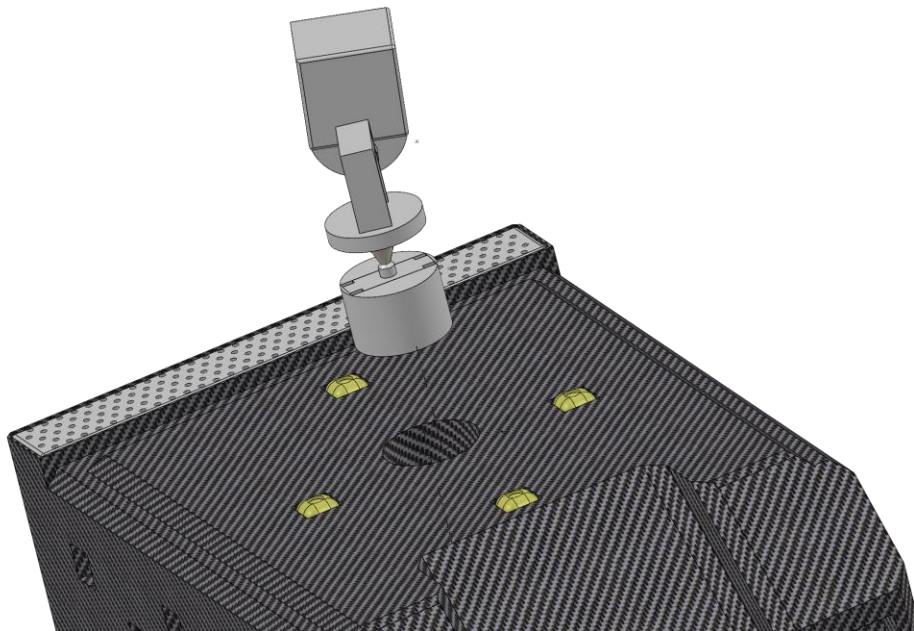


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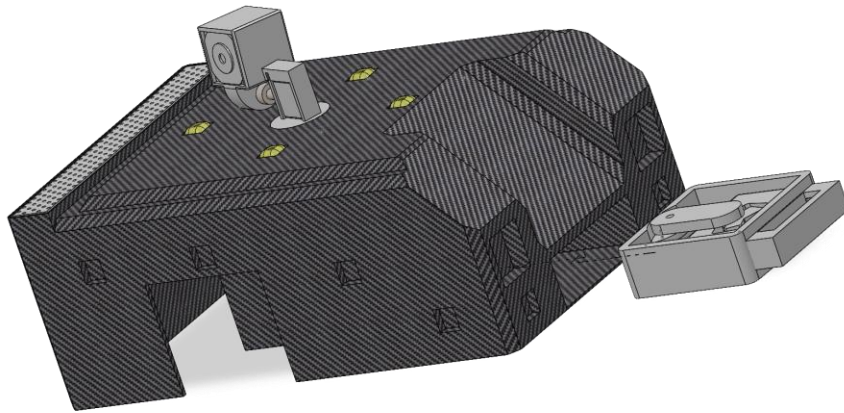


Final Subassembly (Caden Whalen)

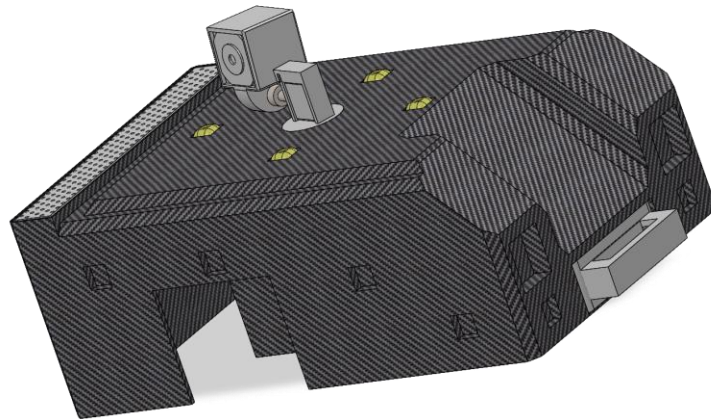
Step 1: Slide the camera subassembly into the circular cutout on top of the frame as shown.



Step 2: Slide the battery subassembly into the rectangular cutout on the back of the frame as shown.

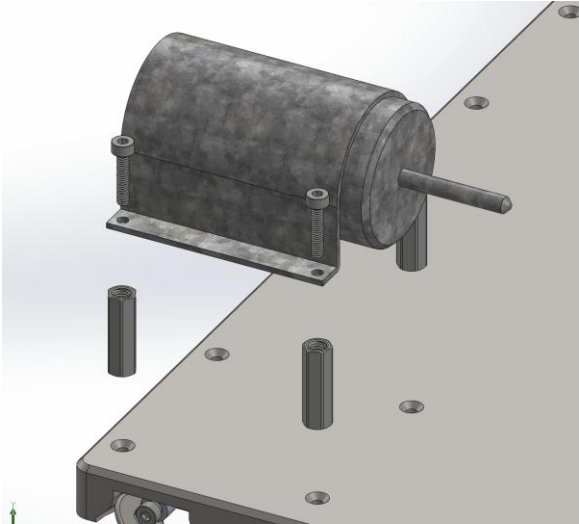


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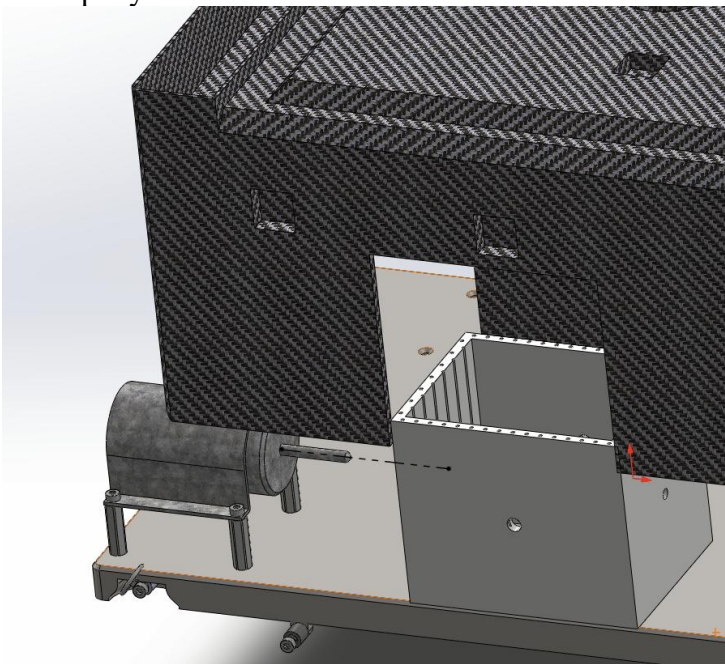
Drivetrain/Suspension Subassembly Instructions (Rayyan Rahman)

Step 1: Using motor mounting bracket, 4x standoffs, and 4x M3 bolts, bolt motor to bottom frame.



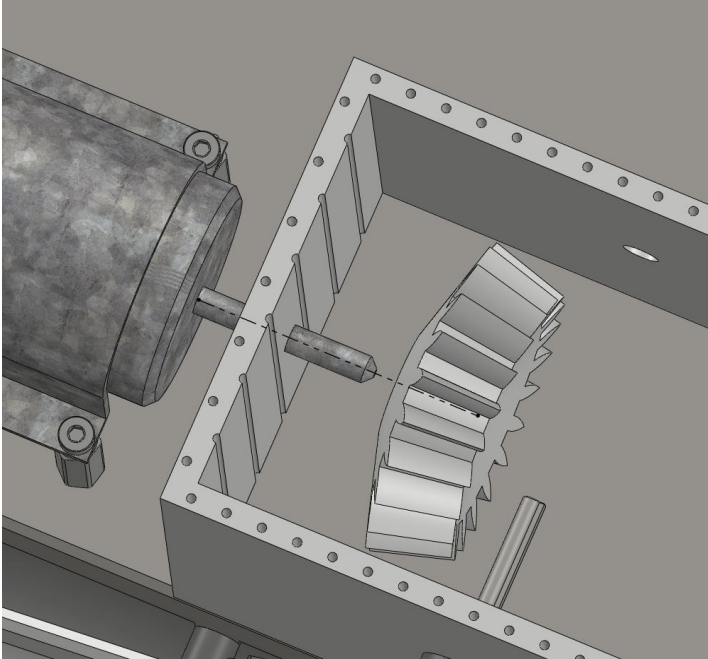
Step 2:

Using the upper frame as a guide (front edge of top frame should be aligned with front edge of back frame), slide the differential box into place, sealing it onto the bottom frame with epoxy.



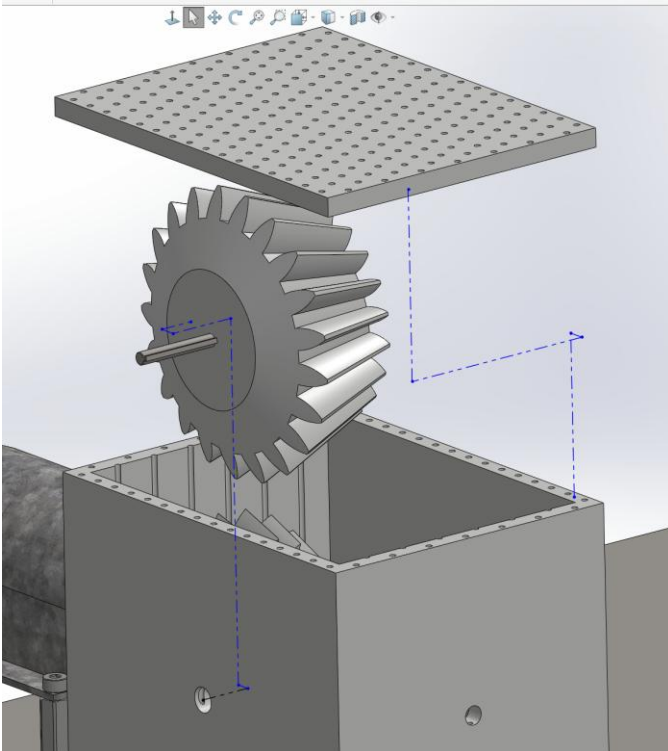
Step 3:

Insert the motor gear (The smaller, thick one) onto the motor. This is an interference fit, so a mallet is necessary. For best function, avoid hitting the edges of the gear.

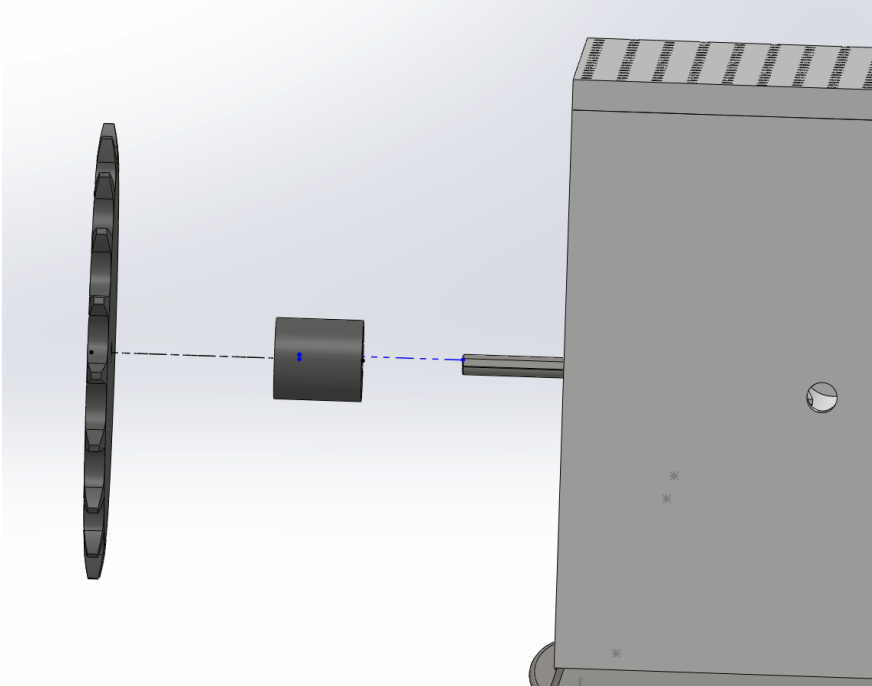


Step 4:

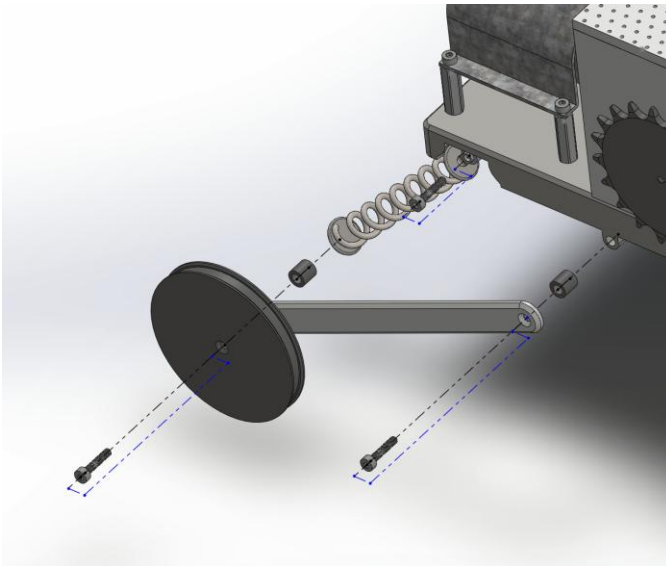
Mallet the keyway into the second sprocket, then put it inside such that the keyway goes through the hole. Then, epoxy the lid of the differential to seal it shut.



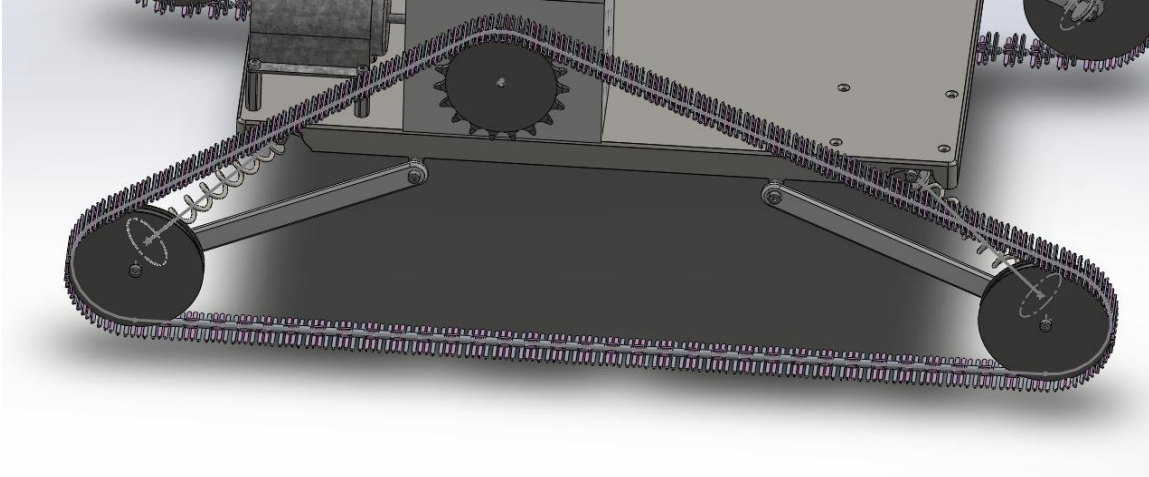
Step 5: Insert the spacer, then press fit on the sprocket.



Step 6: Insert Rocker arm onto frame along with spacer, then align caster, rocker arm, spacer, suspension shock and screw together in that order. Then finally, bolt on shock onto frame. Tighten screw on caster to ensure that caster rolls freely. Repeat on all 4 sides of bottom frame.



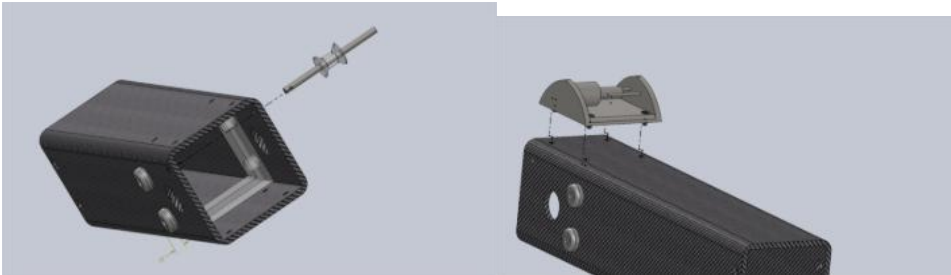
Step 7: Wrap tank treads on sprocket, then around caster. Tank treads are made of tpu, do not be afraid to bend.



Spider Leg Subassembly Instructions (Luke Tumpalan)

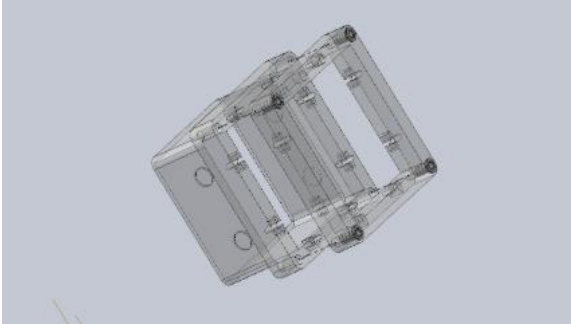
Step 1: Begin assembly with either stage of the telescoping mechanism show is the first stage by inserting the hard stop using 4 M2 thread forming screws

Step 2: Insert shaft with pulley centered then mount motor using 4 M1 thread forming screws



Step 3: Prepare to mount spool onto face plate, secure spool using bushings to center and M1 bolt

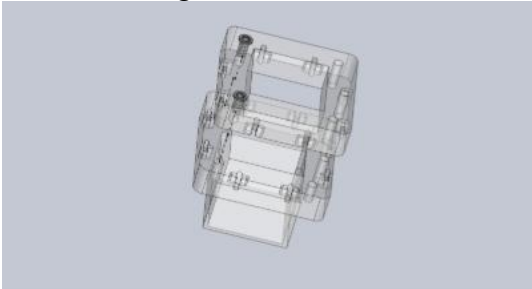
Step4: Using 4 M1 thread forming screws secure face plate assembly together



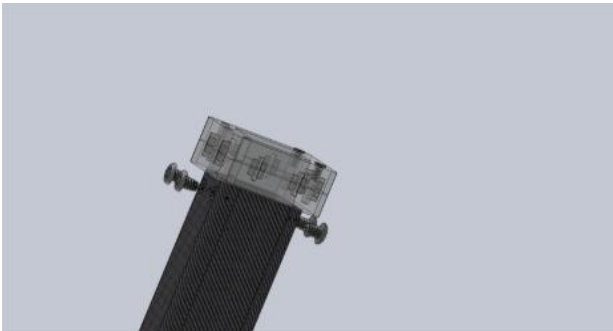
Step 5: Using 2 more M1 thread forming screws secure spool mount to face place and insert faceplate into the first stage to the telescoping portion using 4 M2 thread forming screws



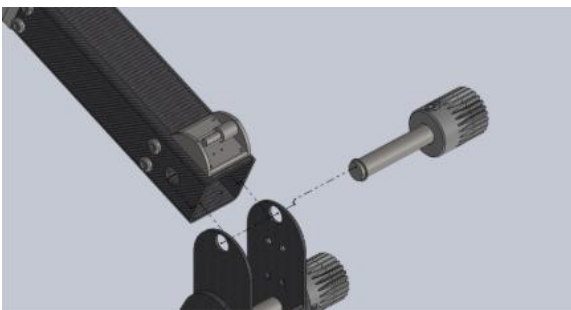
Step 6 : Begin assembly of the second stage with the securing the face plate using 2 M1 thread forming screws



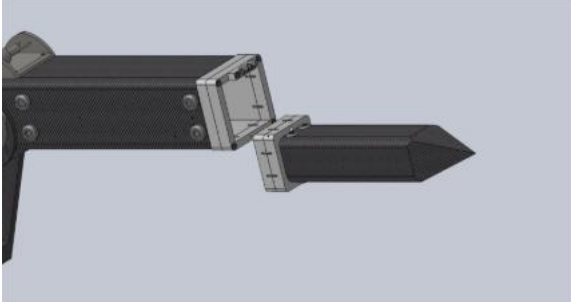
Step 7: Insert face plate into end of the telescoping portion using 4 M2 thread forming screws



Step 8: Connect frame mount to base of telescoping portion, use bushing for spacing between parts on shaft repeat with first stage of telescoping portion to the other side of the base

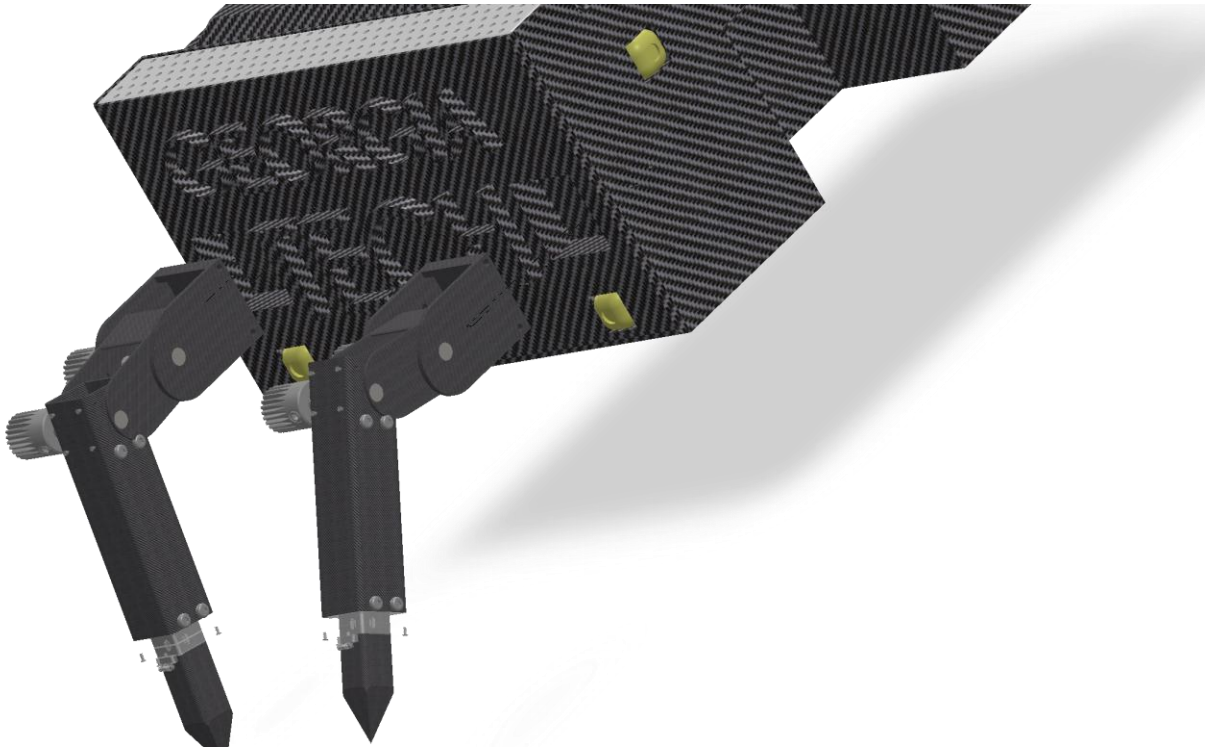


Step 9: Insert last stage into first, run string through that begins and ends in the faceplate on second stage that will tie to the motor mounted on the first stage, run the string around the pulley and spool.

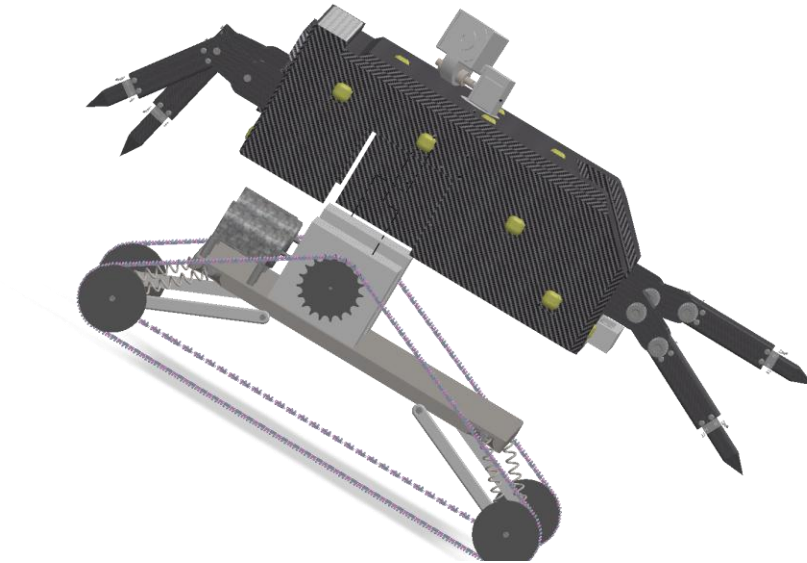


Final Assembly Instructions

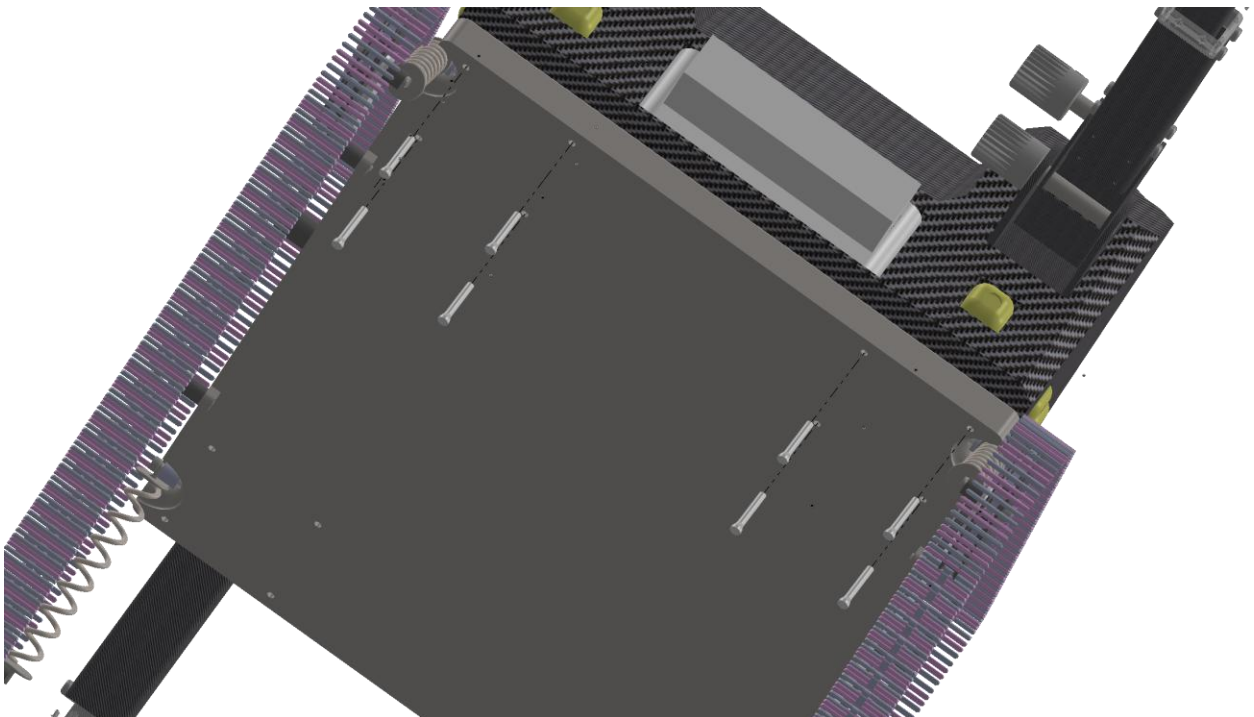
Step 1: Slide the spider arms in the slots located in the front and back of the robot and secure them with structural adhesive bonding.



Step 2: Insert the drivetrain/suspension subassembly into the bottom of the frame as shown.



Step 3: Insert the eight pins into the bottom of the drivetrain/suspension subassembly to secure the system to the frame of the robot.



Final:



Chapter 5

Check for Functionality

Camera Subassembly (Caden Whalen)

Motion 1: The lower servo motor is responsible for the horizontal movement of the camera system. This servo has full range of motion to ensure the camera has a 360-degree view.

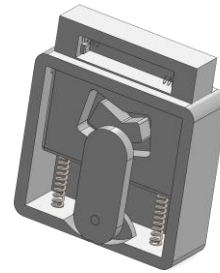
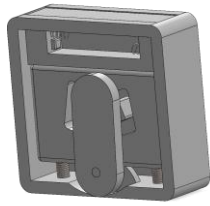


Motion 2: The upper servo motor is responsible for the vertical movement of the camera system. This servo's range of motion is limited to prevent the camera from colliding with the frame of the robot.



Battery Subassembly (Caden Whalen)

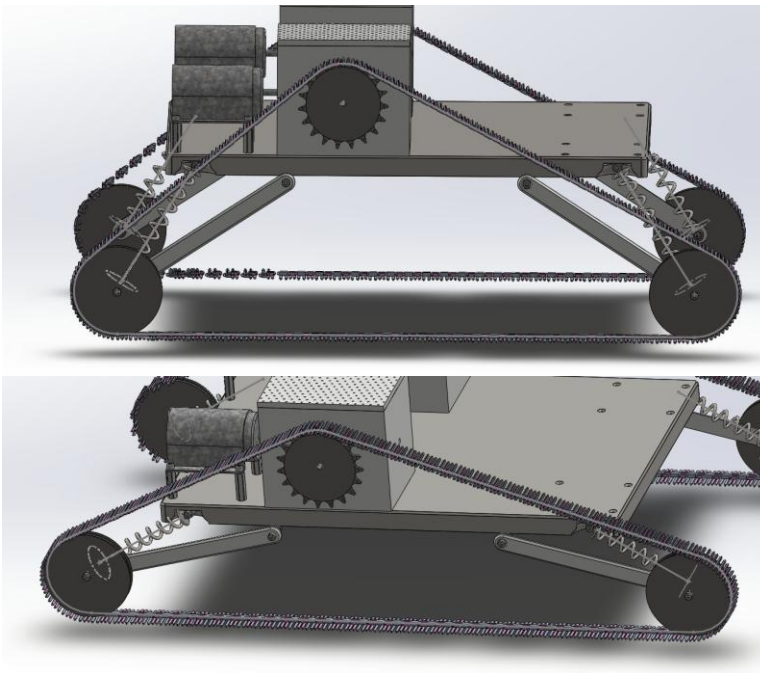
Motion 1: The initial compression of the mobile compartment traps it from moving using the lever attached to the battery frame. The secondary compression frees the mobile compartment from the lever, enabling it to fully extend.



Suspension/Treads Subassembly (Rayyan Rahman)

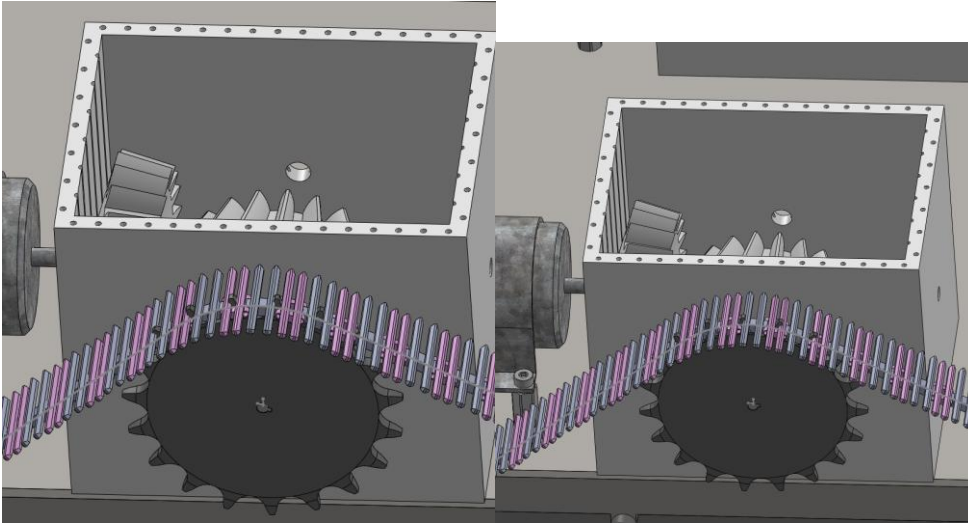
Motion 1:

The compression of the spring moves the tank treads further outwards and increases tension in the treads



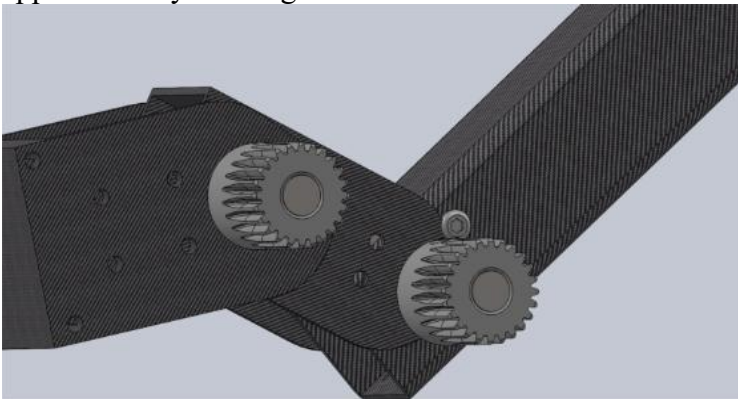
Motion 2:

The motor turns the gears in the differential, turning the sprocket, moving the chain.



Spider Leg Subassembly (Luke Tumpalan)

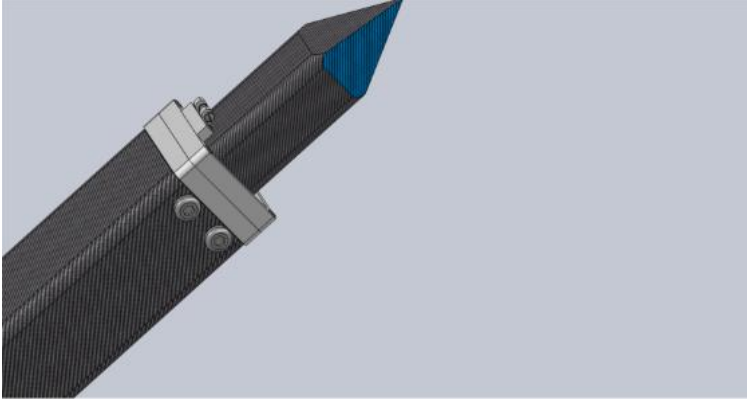
Motion 1: The first joint at connected at the frame mount will allow the leg to rotate approximately 270 degrees



Motion 2: The second joint powered by a direct motor will allow the telescoping portion an extra about 270 degrees of motion



Motion 3: The end of the legs telescope using a string tied to the shaft of the motor mounted on the outside base of the first stage that is tied that wraps around to the top spool and inside pulley where it ends and is tied in the faceplate of the second stage to the telescoping assembly allowing around a 35mm extension



Chapter 6

Summary and Concluding Remarks

When starting our project, our team carefully reviewed the wide range of capstone projects and ended up choosing the Mini Ant Robot for SAR. We chose this capstone project because it looked extremely interesting and quite challenging to design. Up to the challenge, we needed to create a compact robot that could relay visual information to its user while easily maneuvering through rough terrain and tight space to aid in the success search-and-rescue missions. After finishing the team project proposal, we got straight to work on the four main subassemblies that would make our ideas become a functional design that could be used in the real world.

The beginning of our design process heavily revolved around sketching and dimensioning. Before moving onto SolidWorks, each member was responsible for drawing the designs of at least ten complex parts and including appropriate dimensions. Communication was essential in the early stages of design because we had to make sure that each part would fit into their respected subassembly and each subassembly would fit into the final assembly. This extensive planning made the transition to SolidWorks smooth and efficient.

Once in SolidWorks, each member was responsible for creating and ensuring the functionality of their subassembly. Caden was tasked with creating a 360-degree camera system that could relay visual information and a hot swappable battery system involving a push-push mechanism. Rayyan was responsible for the suspension and treads of the robot which enables it to traverse through tight space and protect the electronics held inside. Luke was responsible for creating a system of four spider legs that could help the robot get unstuck in the rubble with a vertical movement. Once done with the subassemblies, we simply used the appropriate mates to ensure that all parts with adequately with the frame.

Overall, the main goals of our Mini Ant Robot for SAR were met with the final design. Our robot is more than capable of recording and relaying information to its user while also having credible mobility in both the vertical and horizontal dimensions of rough terrain. With this being said, we had to make some major compromises and revisions with our robot. The frame of the robot had to be dramatically altered to reduce material usage and ensure functionality. Additionally, the size of our robot's treads had to be increased for the robot to maintain stability while traversing rough terrain.

We had a great time working as a team to create a complex and functional Mini Ant Robot capable of aiding in search-and-rescue missions. Looking back, the only suggestion we could come up regarding this project is to enable teams to collaborate with one another during a designated class.